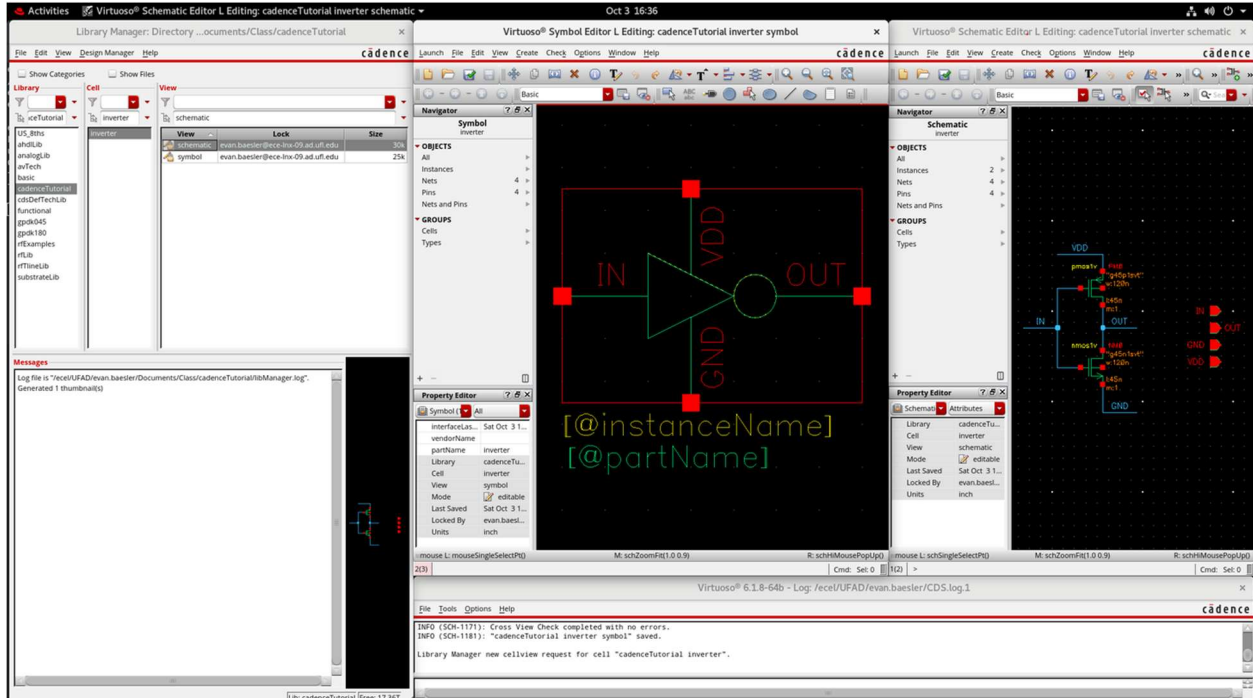


VLSI 1: CADENCE TUTORIAL PT. 1

SCHEMATIC & SYMBOL:



QUESTIONS:

1. What is a Process Design Kit (PDK) and how is it used?
 - a. A PDK is a group of design rules, assets and verification tools provided by a manufacturer to create ICs. They are important because they can determine the manufacturability of a circuit created in Cadence, and provide the basic components needed to make anything.
2. Discuss technology node feature size and how it is defined.
 - a. A technology node feature size is effectively the minimum size that any feature must be to be fabricated and is used to name a generation of technology. A 45nm technology node refers to features on the order of 45nm, but does not guarantee that 45nm is the actual minimum physical feature size.
3. List at least four major foundries (foundries that tape out the chip, not electronic sellers).
 - a. TSMC, Samsung Foundry, GlobalFoundries, and UMC.